

On the cyclic encoding of control₂

Semantic Control Systems (1995), Definition 28

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September 2026

From the paper. Definition 28 (p. 103): “A control₁ system $CS = \langle C, O \rangle$ is also a control₂ system if the variation in the constraint₁ on O , induced by C ’s variation, is itself constrained₂.” Proposition 29 (p. 106) splits the object of control, $O = \langle O_E, O_I \rangle$, giving

$$CS = \langle C, O = \langle O_E, O_I \rangle \rangle \quad (30)$$

and §3.5 (p. 107) names the relations among the three components (Figure 1): “Thus together O_E and O_I form an afferent-efferent loop.”

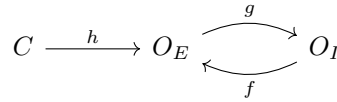
This note does one thing to (30): it records the three named relations of Figure 1 as the arrows of a quiver and takes the free category on it (§1). A theorem follows from the loop alone (§2). §3 says what the theorem does not claim. §4 asks whether the drawing is faithful to what you meant. §5 is a separate provenance item.

1 The encoding

Definition 1 (the quiver Q_J). *Vertices* $\{C, O_E, O_I\}$. *Generating arrows*

$$h: C \rightarrow O_E, \quad g: O_E \rightarrow O_I, \quad f: O_I \rightarrow O_E.$$

Convention. *Arrows point in the functional direction of J95 Figure 1 (p. 107): an arrow $x \rightarrow y$ means “ x acts on y .”*
Read aloud: C disturbs O_E ; O_E constrains O_I ; O_I feeds back to O_E .



symbol	role in Q_J	J95 (page)
C	vertex	“the global environment” (107); “a source of variation to the system O ” (107)
O_E	vertex	“some efferent component of O , the mechanism whose actions assert compensating variation on O_I ” (107)
O_I	vertex	“the controlled variables” (107)
h	arrow	“The variation of C affects O through the relation h ” (106)
g	arrow	“the final constraint ₁ of O_E on O_I ” (106)
f	arrow	“whatever relation there might be back from O_I to O_E ” (106)

Definition 2 (the shape). $\mathcal{I}_J := \mathbf{Paths}(Q_J)$, the free category on Q_J : *objects are the vertices, morphisms are finite directed paths, composition is concatenation, identities are empty paths. No relations are imposed.*¹

Remark 1 (what a quiver forgets). Q_J records only that a named dependency exists. It does not carry: (a) that f, g, h are functions rather than relations (J95 §4.3, p. 114: “usually considered to be functions”); (b) the asymmetry of the loop (§4.3, p. 114: f and g “are not thereby symmetric or complementary”); (c) the hierarchy of C over O and the “large loop gain” (p. 108); (d) the second loop. §3.7.2 (p. 110) says “there is also a kind of loop between C and O_E ” by reciprocal variation, and §3.7.1 (p. 109) rules it one-way at the level of action (“ O_E mirror the actions of C , but not vice versa”). Q_J follows §3.7.1: h is a single arrow. (e) Turchin’s alternative, in which g “flows only through the environment” (p. 108). Items (d) and (e) are encoding forks, not oversights; each yields a different Q .

2 The theorem

Write $\ell := f \circ g \in \mathbf{Hom}_{\mathcal{I}_J}(O_E, O_E)$ for the feedback loop, a path of length 2.²

Lemma 1. *For $n \in \mathbb{N}$, ℓ^n has length $2n$; hence $n \mapsto \ell^n$ is injective and $\mathbf{Hom}_{\mathcal{I}_J}(O_E, O_E)$ is infinite.*³

Proof. Length is additive under concatenation, so $|\ell^n| = n|\ell| = 2n$. Distinct n give distinct lengths, hence distinct paths. $\square$

*J95 = C. Joslyn, *World Futures* 45 (1995) 87–123. Lean: `systems-science-foundations/Systems/Category/{ShapeJoslyn, CyclicObstruction, JoslynIncomparability}.lean`, checkout 4288793. Drafted with Claude (Table 5.1); the encoding decision, the readings of J95, and §4 are mine.

¹JoslynShape := Paths JoslynPosition, ShapeJoslyn.lean. Paths and Path.length.comp are Mathlib’s.

²joslynLoop, joslynLoop.length : joslynLoop.length = 2 := rfl.

³cyclePow.length, cyclePow.injective, infinite.path.of.cycle, all proved for an arbitrary quiver and an arbitrary positive-length loop.

Theorem 1. *Let $\mathcal{C}$ be any category in which every hom-set is finite. No functor $F: \mathcal{I}_J \rightarrow \mathcal{C}$ is faithful.*⁴

Proof. A faithful functor is injective on each hom-set. By Lemma 1 it would inject the infinite set $\text{Hom}(O_E, O_E)$ into the finite set $\text{Hom}(FO_E, FO_E)$. $\square$

Corollary 1. *Let $\mathcal{I}_K := \text{Paths}(R \rightarrow T)$ be the shape of Klir’s $S = (T, R)$: two objects, one generating arrow. Every hom-set of $\mathcal{I}_K$ is a subsingleton, hence finite. There is no faithful functor $\mathcal{I}_J \rightarrow \mathcal{I}_K$.*⁵

Remark 2 (why this is the boundary and not a curiosity). Every other tradition encoded in the same program (Klir, Bunge, Mobus, Myers, Wymore, Mesarović) has an acyclic Q , hence finite hom-sets, and is placed relative to the others by faithful comparison functors. Q_J is the first cyclic shape in the landscape; the only other is Spivak’s value loop, which instantiates the same lemma. Feedback is where the finite-shape method stops; Theorem 1 makes that folklore a theorem whose sole premise is the cycle.

3 What the theorem does not say

1. **It is presentation-relative.** From the theorem’s own docstring, written before anyone asked you: “both shapes are encodings of primary texts as dependency quivers, and a different defensible encoding of Joslyn 1995 — one that does not make the control loop a cycle at quiver level — could dissolve the obstruction. Not a maximality result.”
2. **It is about Paths, which imposes no relations.** Impose one and it can die. If feedback is taken to be idempotent, $\ell \circ \ell = \ell$, then $\text{Hom}(O_E, O_E)$ collapses to $\{\text{id}, \ell\}$ and Theorem 1 fails. Whether control_2 licenses that identification is a claim about your text, not about category theory.
3. **Re-bracketing does not dissolve it.** Your (31) $\langle O_E, \langle C, O_I \rangle \rangle$ and (32) $\langle O_I, \langle C, O_E \rangle \rangle$ regroup the vertices; neither removes an arrow, so the loop and the theorem survive both. (32) adds a second loop if the “kind of loop” of §3.7.2 is drawn, which only strengthens the obstruction.
4. **Unrolling does dissolve it.** Index the loop by level: vertices $C, O_E^{(k)}, O_I^{(k)}$ for $k \in \mathbb{N}$; arrows $h: C \rightarrow O_E^{(0)}, g_k: O_E^{(k)} \rightarrow O_I^{(k)}, f_k: O_I^{(k)} \rightarrow O_E^{(k+1)}$. This Q is acyclic, every hom-set is finite (at most one path between any two vertices), and faithful functors into $\mathcal{I}_K$ are no longer excluded on cardinality grounds. *Stated, not machine-checked.*
5. **It does not say control_2 is incomparable in principle.** It says the method that places the six acyclic traditions cannot reach a cyclic one. A quotient, an enrichment, or a 2-categorical treatment of the loop is neither ruled out nor attempted here.

4 The question, which only you can answer

In the margin of the Summer 2025 paper (p. 4) you wrote: “One could argue that control is always so circular.” The encoding took that literally, at quiver level, in one system, with no relations on the loop’s powers. Items 2 through 4 of §3 are then one question: *where does the closure of control_2 live?* In the vocabulary of your 2026 abstract: is the semantic closure level-flat, so that each ℓ^n is a genuinely distinct control trajectory, or does it unroll across a metasystem transition, so that the cycle is an artefact of drawing every level on one page? Three outcomes, each a result:

1. **You defend the cycle.** Theorem 1 becomes author-endorsed, the strongest provenance a formalization can carry.
2. **You propose an acyclic encoding** (the unrolling of §3 item 4, or another). The obstruction dissolves for that encoding; presentation-relativity is exhibited rather than asserted.
3. **You rule both encodings defensible.** One text, two shapes, different comparison behaviour: the sharpest exhibit yet for the layer of validation no proof checker sees.

5 One provenance item, separately

Definition 28 is entry 009 of the definition atlas at `math.systems`, page-image transcribed and hand-checked, with five of your case rulings (thermostat and the decoupled-envelope pair admitted; damped spring, boiling liquid, spinning top refused). One grading is offered for correction. Definition 25 (control_1) is introduced on p. 101 as Turchin’s 1992 definition, “we offer it again here,” so the catalogue grades it a restatement of Turchin and records Definition 28 as the first control definition that is yours alone. If that reading is wrong, the correction is exactly what the catalogue exists to receive.

⁴`joslyn.no_faithful_functor`, instance of the general `paths.no_faithful_functor_of_cycle`. Axioms: `propext`, `Classical.choice`, `Quot.sound`. Oh sorry.

⁵`no_faithful_joslyn_to_klir`; the finiteness hypothesis is discharged by an instance built from `klirHomSubsingleton`, not by inspection in prose.